

Networking

Objective

This roadmap is designed to build a strong networking foundation,

Level 1 – Foundation

Topics

- ❖ OSI Model
- ❖ TCP/IP Model
- ❖ IP Addressing (IPv4)
- ❖ Private vs Public IP

Learning Outcomes

By the end of Day 1, you will be able to:

- ✓ Explain the OSI Model and its seven layers.
 - ✓ Differentiate between the OSI and TCP/IP models.
 - ✓ Understand IPv4 addressing.
 - ✓ Distinguish between private and public IP addresses.
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Level 2 – IP Mastery

Topics

- ❖ Subnetting
- ❖ CIDR
- ❖ Network ID
- ❖ Broadcast ID
- ❖ Practice Questions (15–20)

Learning Outcomes

By the end of Day 2, you will be able to:

- ✓ Perform subnet calculations.
- ✓ Understand CIDR notation.

- ✓ Identify Network IDs and Broadcast IDs.
 - ✓ Solve interview-style subnetting questions.
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Level 3 – Network Services

Topics

- ❖ DNS
- ❖ DHCP
- ❖ ARP
- ❖ NAT

Learning Outcomes

By the end of Day 3, you will be able to:

- ✓ Explain how DNS resolves domain names.
 - ✓ Understand IP assignment using DHCP.
 - ✓ Describe ARP and MAC address resolution.
 - ✓ Explain how NAT enables private networks to access the Internet.
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Day 4 – Data Movement

Topics

- ❖ Routing Basics
- ❖ Switching Basics
- ❖ VLAN

Learning Outcomes

By the end of Day 4, you will be able to:

- ✓ Explain how routers forward packets.
 - ✓ Understand the role of switches in LANs.
 - ✓ Configure and explain VLAN concepts at a high level.
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Level 5 – DevOps Networking

Topics

- ❖ Ports & Protocols
- ❖ HTTP / HTTPS
- ❖ SSH
- ❖ End-to-End Request Flow (Browser → DNS → Server → Response)

Learning Outcomes

By the end of Day 5, you will be able to:

- ✓ Identify common ports and protocols.
- ✓ Explain the difference between HTTP and HTTPS.
- ✓ Use SSH for secure remote access.
- ✓ Describe the complete lifecycle of a web request from browser to server and back.